Multi-Dialect Marathi Text Normalization via LLM-Driven Synthetic Augmentation

Aditya Kinjawadekar, Surender Kannaiyan, and Saugata Sinha

Visvesvaraya National Institute of Technology, Nagpur, India
kinjawadekaradi112@gmail.com

Abstract. Keywords: Dialect Normalization · Low-Resource NLP · Marathi Dialects · Sequence-to-Sequence Modeling · Synthetic Data Augmentation · mT5 · IndicBART · Text Standardization.

1 Introduction

Rapid development of deep learning architectures and self-attention mechanisms [1] have made it possible to develop systems capable of understanding and generating natural language. Open-weight models are used in processing text across various domains involving contextual data as they allow finer control over the model, environment and the data [2,3]. Even with the given advancements , current systems involving such models tend to cater requests towards standard languages showing undesired output for queries with non-standard dialects [4]. Various surveys conducted [5,6] have proven that speakers with native dialect have faced several issues within the existing language processing systems such as Automatic Speech Recognition and Machine Translation due to lexical and phonetic variations [7,8].

Dialect normalization is the task of transducing regional dialectal text into standard written language, where the main difficulty lies in keeping the original meaning intact [8]. It is an important layer for digital linguistic inclusion. As countries like India have 22 officially recognized languages including hundreds of regional dialects spoken by 1.4 billion people, sub-regional varieties have remained unexplored when it comes to computational research. An example of such linguistic disparity is Marathi, an Indo-Aryan language spoken by over 83 million people in western India [9]. In this work, our core motivation stems from analyzing what drives dialectal disparities and why standard models underperform on regional text. We hypothesized that evaluating multi-dialect Marathi text as a single homogenous benchmark masks critical sub-regional nuances. Therefore, we systematically investigate how different regional dialects (Southern Konkan, Northern Konkan, and Varhadi) behave under normalization, proving that evaluation metrics vary significantly across dialects due to distinct lexical, phonetic, and morpho-syntactic properties.

Building pipelines for translating Indic dialects primarily comes down to two core challenges:

1. **Extreme Data Scarcity**: Datasets that map regional dialects to standard forms are technically non-existent; and
2. **Morphological Complexity**: Marathi is a language with rich morphology, where noun cases, postpositional affixes, and verbal tense suffixes exhibit complex variations [10, 11].

Addressing these limitations, this work presents a multi-dialect Marathi text normalization framework combining synthetic data augmentation with sequence-to-sequence transformer fine-tuning. To overcome severe data scarcity while ensuring that synthetic augmentation yields high accuracy of results without introducing noise, we implement an automated multi-tier verification engine (combining script filtering and LLM semantic auditing). Furthermore, we rigorously evaluate the performance of existing pre-trained multilingual sequence-to-sequence models—specifically google/mT5-small (300M) and IndicBART (244M)—to benchmark their capability in capturing complex dialectal shifts.

Our core contributions are:

- **Parallel Corpus Generation**: Based on the transcripts of RESPIN dataset [12], we construct a parallel corpus of three Marathi dialects (Southern Konkan, Northern Konkan, and Varhadi) mapped to Standard Marathi across agricultural and banking domains.
- **Automated Synthetic Augmentation with Verification**: We implement a synthetic data expansion pipeline using Gemma-4 [3] with a verification step that flags and removes corrupted pairs, proving that verified synthetic augmentation consistently provides higher accuracy of results.
- **Cross-Dialect & Multilingual Model Benchmarks**: We provide a comprehensive empirical evaluation of existing multilingual models (mT5-small vs. IndicBART), demonstrating that evaluation results vary significantly across individual dialects (e.g., up to 80.99 BLEU on Varhadi) and establishing benchmark standards for Indic dialect normalization.

2 Related Work

2.1 Regional Dialect Standardization across Language Families

Early approaches in the context of European languages used Character-Level Statistical Machine Translation (CSMT) and phrase-based models for Swiss German Dialects [13] which were later replaced by pre-trained language models [14, 15]. Similar approaches have been developed for Finnish, Swedish, and Estonian dialects [11, 16, 17]. A study of four language families by Kuparinen et al. [8] suggested that fine-tuned sequence-to-sequence transformers surpassed the rule-based and SMT baselines. Other than European countries, East Asian and African-Asian varieties have also shown similar patterns. Japanese regional dialects have been processed using phrase-level LSTM modules [18], while Mandarin scripts relied on acoustic-phonetic mapping [19]. The Conventional Orthography for Dialectal Arabic (CODA) framework [20]in the landscape of Arabic languages gave guidelines for Modern Standard Arabic (MSA) which led to multi-city translation benchmarks (MADAR [21], NADI [22]) and morphology-oriented prompt design for LLMs [23, 24].

2.2 Indic Dialect Landscape and Marathi NLP

Even though pre-trained models such as IndicBART [9] and mT5 [25]which are trained on multilingual corpora cover major languages, they lack understanding of regional sub-dialects. Models based on a single language like L3Cube-MahaNLP [26] offer pre-trained marathi models for classification, but do not support generative text standardization. INDIC-DIALECT [27] considers Hindi varieties (Bhojpuri, Magahi, Maithili),for evaluation via WFST frameworks for text normalization while preserving the underlying semantics [28]. However, Marathi dialects were primarily dealt for acoustic/ASR classification [29, 30], or zero-shot LLM retrieval [31, 32]. Our work fills this gap by presenting an end-to-end neural fine-tuning pipeline with evaluations and data augmentation strategies.

2.3 LLM Distillation and Multi-Tier Verification

LLMs can be used to suffice for data scarcity in various tasks. [33–35]. Knowledge distillation strategies can be employed to train compact and efficient models based on the outputs generated by a larger teacher model. [36–38]. But the use of such techniques for low-resource regional dialects involves significant amount of hallucinations and noise. [38, 39]. We solve this by utilising Gemma-4 with a robust morphologically aware system prompt and a multi-tier verification system that ensures integrity of the data generated.

3 Methodology

3.1 Target Dialects and Linguistic Properties

The following Marathi dialects based on RESPIN corpus were on our radar that exhibit unique lexical and phonetic variations:

1. **Southern Konkan**: Shortening of final-word vowels, softening of dental consonants, and different auxiliary verb forms are its key characteristics. Lexical differences include vowel shifts: standard जातो ("he goes") becomes जातां; standard कुठे ("where") becomes कुठंय.
2. **Northern Konkan**:Since it is surrounded by neighboring Gujarati and Bhili language families,it tends to exhibit significant consonant variations. Few examples are: standard कशाला ("why") → Northern Konkan कसाले; standard मुलगा ("boy") → Northern Konkan पोर.
3. **Varhadi**: Spoken in the regions of Vidarbha accompanied with verb suffix modifications. Standard questioning का? ("why?") becomes काऊन?; verbal forms exhibit noticable patterns. Varhadi maintains closest structural similarity to Standard Marathi as compared to other dialects since the transformations are more systematic.

Table 1. RESPIN-S1.0 Marathi Source Corpus Statistics

Code	Dialect Name	Utterances (%)	Duration (h)	Speakers	Unique Prompts
D1	Southern Konkan (Malvani)	203,651 (25.14%)	247.31	732	5,838
D2	Northern Konkan (Ahirani)	198,560 (24.52%)	265.25	557	5,825
D3	Standard Marathi (Puneri)	208,850 (25.79%)	255.79	608	6,077
D4	Varhadi (Vidarbha)	198,873 (24.55%)	257.70	747	5,330
Total	**All Dialects**	**809,934 (100%)**	**1,026.06**	**2,644**	**23,069**

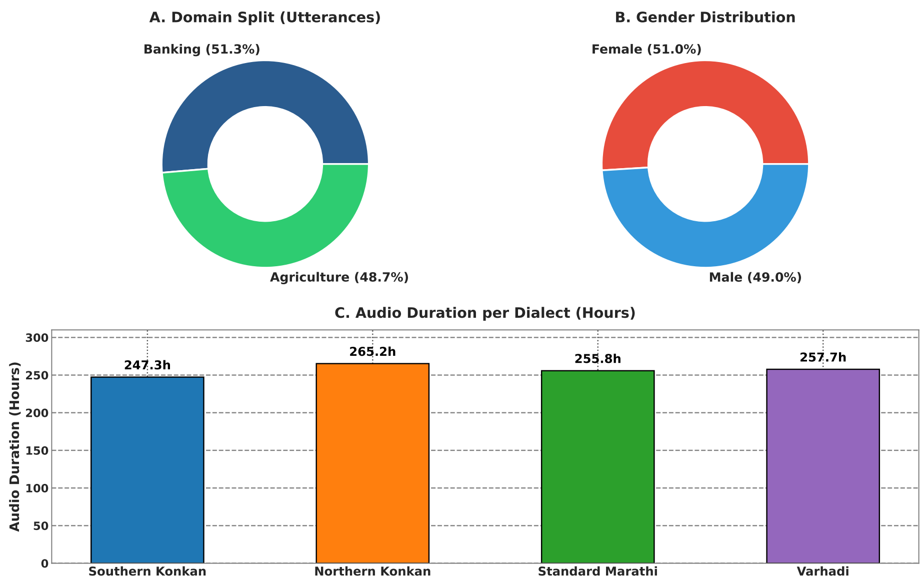

Fig. 1. RESPIN-S1.0 Marathi source corpus demographics: (A) domain distribution ; (B) gender split ; (C) audio recording duration per dialect variant.

3.2 Original Corpus Curation and Foundation in RESPIN

Our primary corpus still remains the domain-specific transcripts of the RESPIN-S1.0 initiative [12], developed by researchers at IISc Bangalore. RESPIN has audio recordings and their transcripts around agriculture and banking domains for different Indian languages and their dialects, it still lacks aligned dialect-to-target language translations which is needed for normalisation.

3.3 LLM-Driven Corpus Generation and Verification

We considered several design decisions to ensure generation quality to be maximised: To deal with the unavaiability of data we developed a synthetic data generation pipeline considering a few design aspects in order to ensure the maximum quality of data generation:

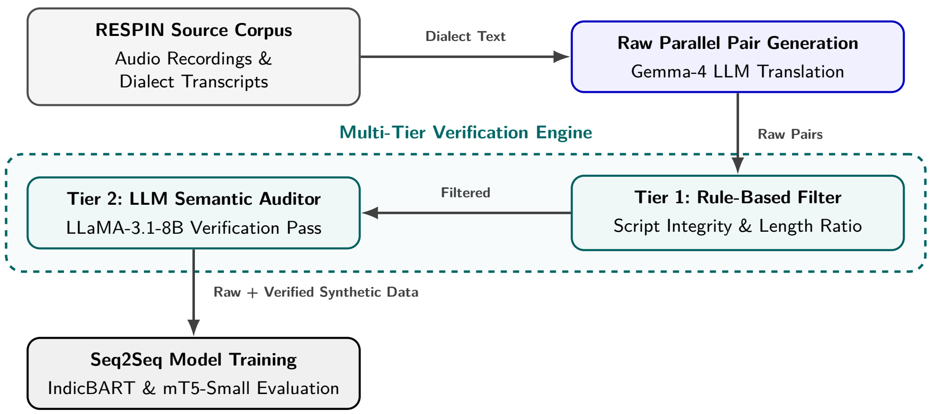

Fig. 2. System architecture of the multi-dialect normalization framework.

- **1-Prompt-Per-Sentence Architecture**: Instead of batching multiple pairs per API call (which risks contextual loss), each sentence is processed separately.
- **Detailed Prompting Strategy**: System prompt is designed for the output to preserve exact meaning, context regarding the domain, proper nouns, and numerical values from the input with a set of 5-8 few-shot examples per dialect.
- **Temperature Control**: Generation temperature is set to 0.3 with top-p = 0.9 for optimal results.

This set was generated using Gemma-4 [3] under the 12B configuration accessed via Ollama, run locally. All generated pairs undergo a three-tier verification and correction pipeline:

1. **Tier 1: Rule-Based Syntactic Filter.**
   - *Devanagari Script Integrity*: Both source and target must have at least 90% Devanagari Unicode characters that lie in the `U+0900--U+097F` range, that automatically get rid of unsolicited characters.
   - *Length Ratio Constraint*: $0.5 \leq |L_{\text{dialect}}|/|L_{\text{standard}}| \leq 2.0$ to avoid hallucinations.
   - *Non-Identity Constraint*: $L_{\text{dialect}} \neq L_{\text{standard}}$ to flag identical string copies.
   - *Minimum Token Count*: Both source and target must contain at least 3 whitespace-separated tokens.
2. **Tier 2: LLM Semantic Verification.** A secondary round of evaluation was performed using `llama-3.1-8b-instant` (accessed via Groq API). The verification happens on a scale of 1-5 score that considers semantics, hallucinations , multilingual contamination; and complete standardization of dialects. Pairs that score < 4.0 are rejected or routed for further correction.

Table 2. Statistics based on generated and verified data

Dialect Name	Original Clean Pairs	Raw Synthetic Generated	Clean Synthetic Verified	Corrupted / Rejected Pairs (%)	Total Clean Expanded Pairs
Southern Konkan	5,576	5,838	5,569	269 (4.61%)	**11,145**
Northern Konkan	5,501	5,825	5,534	291 (5.00%)	**11,035**
Varhadi	5,086	5,330	5,069	261 (4.90%)	**10,155**
Total	**16,163**	**16,993**	**16,172**	**821 (4.83%)**	**32,335**

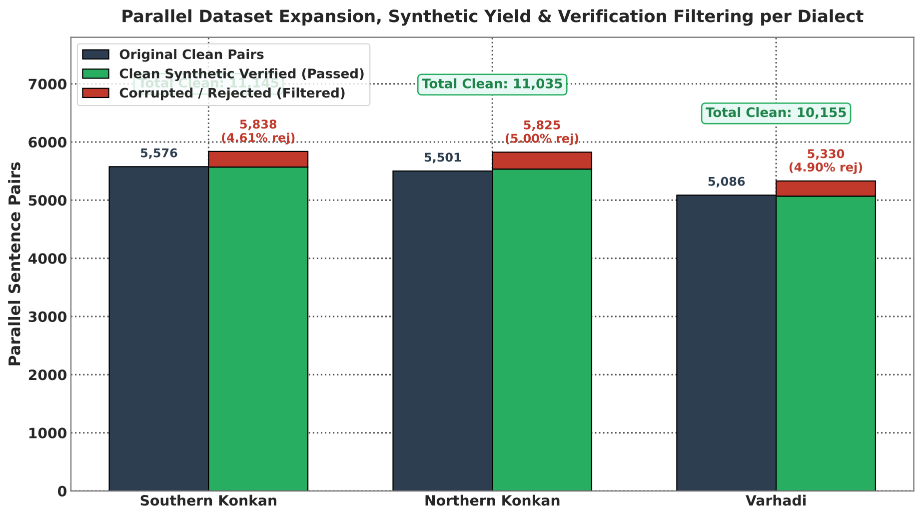

Fig. 3. Parallel dataset expansion composition per dialect partition, comparing original parallel pairs against synthetic verified pairs produced by the LLM generation and auditing pipeline.

3. **Closed-Loop Correction Engine.** Pairs that are flagged by Tier 1 or Tier 2 enter a correction loop: Step 1 generates corrected translations using specialised prompts based on pre-defined reason of identified failure (e.g., “Hindi leakage detected” or “Incomplete normalization”); Step 2 verifies the corrected sentences again,following the complete discard of pairs that fail the second audit pass.

The given pipeline was used to generate parellel pairs from the RESPIN transcripts. The same pipeline was used further for generation of additional synthetic data for expansion. Out of the 16,993 synthetic parallel pairs generated by the LLM pipeline, our multi-tier verification engine flagged and discarded 821 flawed pairs (an overall corruption/rejection rate of 4.83%), yielding 16,172 clean verified pairs as mentioned in Table 2. Combining with the 16,163 clean original pairs, the final expanded dataset consists of 32,335 verified parallel sentence pairs across all three dialects.

Table 3. Comprehensive Evaluation Matrix on `IISc_RESPIN_test_mr` Benchmark Set (2,170 Utterances)

Model Scope	Training Data	Dialect / Subset	Utts	IndicBART (244M)		mT5-Small (300M)	
				BLEU (↑)	WER (↓)	BLEU (↑)	WER (↓)
Single-Dialect	Original	Southern Konkan	559	**57.80**	**26.60%**	43.86	34.37%
		Northern Konkan	540	**90.13**	**6.46%**	79.46	11.95%
		Varhadi	516	**83.59**	**10.19%**	74.81	14.73%
	Synthetically Expanded	Southern Konkan	559	24.06	60.32%	**44.36**	**32.75%**
		Northern Konkan	540	65.41	28.70%	**79.76**	**12.61%**
		Varhadi	516	67.97	25.43%	**76.90**	**14.19%**
Multi-Dialect	Original	Southern Konkan	559	26.21	53.15%	**40.94**	**35.34%**
		Northern Konkan	540	47.59	32.84%	**77.50**	**14.65%**
		Standard Benchmark	555	96.12	3.12%	**96.65**	**1.91%**
		Varhadi	516	72.75	25.30%	**73.47**	**16.16%**
	Synthetically Expanded	Southern Konkan	559	**52.06**	36.80%	43.88	**34.96%**
		Northern Konkan	540	49.08	28.70%	**78.16**	**13.55%**
		Standard Benchmark	555	96.80	1.65%	**97.39**	**1.37%**
		Varhadi	516	70.27	21.40%	**74.74**	**15.57%**

3.4 Evaluation Protocol

For stable performance during development, models were validated using 5-Fold Cross-Validation on an 85/15 stratified split of the train set. The final reports were generated on the official held-out test set provided by RESPIN after training the model on the complete dataset, which had around 2,170 utterances across target dialects. This setup is applied across all dialects on the original transcript-based data and their synthetically expanded versions.e Results are reported using metrics that are evaluated using SacreBLEU [40] and Word Error Rate (WER) scripts:

- **BLEU** (Bilingual Evaluation Understudy): BLEU-4 with 4-gram precision, brevity penalty, with an exponential smoothing method.
- **chrF++**: Character n-gram F-score combined with word unigrams and bi-grams.
- **Normalized WER / CER**: Word Error Rate and Character Error Rate after Devanagari Unicode script standardization.

4 Findings

4.1 Comprehensive Test Benchmark on `IISc_RESPIN_test_mr`

Key observations from Table 3 and Figure 4 include:

1. **mT5-Small Performance**: `mT5-small` upon finetuning on the synthetically expanded dataset achieves 73.48 BLEU and 87.70 chrF++ on the overall test set.
2. **IndicBART Scaling**: Fine-tuning IndicBART on the synthetically expanded multi-dialect corpus achieves **76.50 BLEU** and **88.04 chrF++**, suggesting that larger data size brings up the efficiency in mBART-50 architectures.

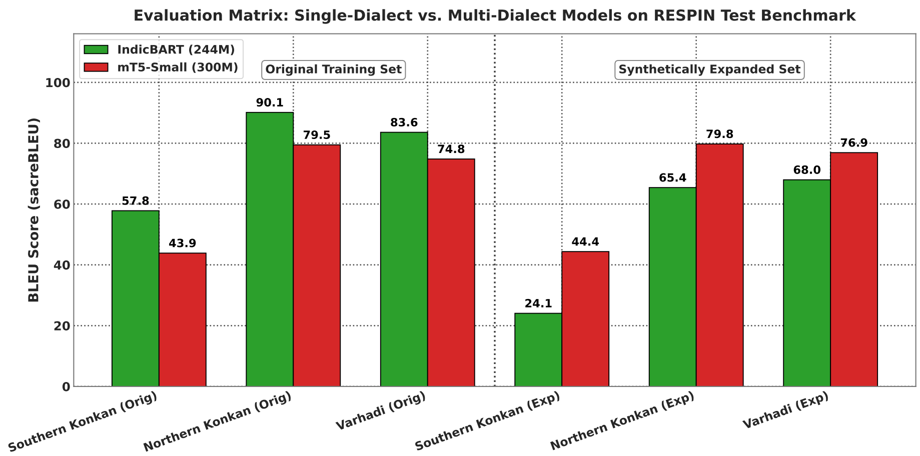

Fig. 4. Comprehensive sequence-to-sequence evaluation matrix on the held-out RESPIN test benchmark (2,170 utterances), comparing IndicBART (244M) and mT5-small (300M) across single-dialect and multi-dialect training configurations.

Table 4. Benchmark Performance Matrix on Original Datasets: **Single-Dialect** Models (5-Fold CV)

Dialect Name	**Total Pairs**	**Test Set (IB / mT5)**	**IndicBART (244M)**		**mT5-Small (300M)**	
			BLEU	**chrF++**	**BLEU**	**chrF++**
Southern Konkan (D1)	5,576	837 / 836	**48.54**	**78.35**	46.31	74.43
Northern Konkan (D2)	5,501	826 / 825	**60.58**	**80.30**	60.31	77.34
Varhadi (D4)	5,086	763 / 762	76.63	90.93	**81.00**	**91.21**

4.2 Single-Dialect vs. Multi-Dialect Benchmark Results on Original and Synthetic Datasets

Table 4 presents the performance matrix on original single-dialect benchmarks, and Table 5 explains the dialect-wise breakdown of the multi-dialect model on original data.

Table 6 presents the performance matrix on synthetically expanded single-dialect benchmarks, and Table 7 details the dialectwise evaluation breakdown of the multi-dialect model on expanded data.

4.3 Impact of Synthetic Data Augmentation

Table 8 isolates the effect of dataset scaling by comparing performance across original and synthetically expanded corpora for each model architecture. Overall, multi-dialect models exhibit significant net gains (+8.95 BLEU for IndicBART, +6.22 BLEU for mT5-small), confirming that data scarcity was a primary performance bottleneck.

Table 5. Benchmark Performance Matrix on Original Datasets: **Multi-Dialect** Dialectwise Breakdown (5-Fold CV)

Evaluated Subset	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkan (D1)	16,163	837 / 836	26.21	53.15	**48.28**	**75.04**
Northern Konkan (D2)	16,163	826 / 798	47.59	67.16	**62.35**	**78.92**
Varhadi (D4)	16,163	763 / 790	72.75	85.63	**79.57**	**90.51**
Overall Combined	**16,163**	**2,426 / 2,424**	**48.17**	**68.58**	**63.29**	**81.37**

Table 6. Benchmark Performance Matrix on Synthetically Expanded Datasets: **Single-Dialect** Models (5-Fold CV)

Dialect Name	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkan (D1)	11,145	1,672 / 1,671	47.25	64.69	**65.10**	**82.88**
Northern Konkan (D2)	11,035	1,656 / 1,655	40.51	62.21	**62.07**	**79.29**
Varhadi (D4)	10,155	1,524 / 1,523	73.62	86.75	**78.89**	**90.59**

But a dialect-wise breakdown suggests notable variance across the corpora. Southern Konkan showed extreme upside due to expansion (+25.85 BLEU for IndicBART, +18.34 BLEU for mT5-small) while Northern Konkan sees decent gains (+1.49 BLEU for IndicBART, +0.37 BLEU for mT5-small) due to its already high original baseline performance. Following are our estimates that could justify the given performance:

1. **Baseline Saturation and Precision Dilution**: Varhadi already had better performance on original non-expanded set (72.75-79.57 BLEU). We estimate that LLM-generated synthetic pairs may introduce suffix rules which were over-generalized that slightly degraded the exact N-gram matching against the original reference texts.
2. **Cross-Dialect Parameter Competition**: In shared dialect training, we estimate that appending synthetic data across distinct sub-dialects may have caused performance competition in shared attention weights, which could have considered prioritizing recovery on severely sparse dialects (Southern Konkan) over saturated ones.

4.4 Verification Engine Impact: Raw Unverified vs. Filtered Clean Data

Performance of our verification and filtering engine is measured by conducting a comparative analysis on models trained on *Raw Unverified Synthetic Data* versus *Filtered Clean Synthetic Data* (the expanded suite). The raw unverified dataset contains 19,914 parallel pairs, that included 3,742 rejected pairs that

Table 7. Benchmark Performance Matrix on Synthetically Expanded Datasets: **Multi-Dialect** Dialectwise Breakdown (5-Fold CV)

Evaluated Subset	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkan (D1)	32,335	1,672 / 1,710	52.06	72.15	**66.62**	**83.57**
Northern Konkan (D2)	32,335	1,656 / 1,627	49.08	69.90	**62.72**	**79.63**
Varhadi (D4)	32,335	1,524 / 1,513	70.27	84.27	**78.73**	**90.46**
Overall Combined	**32,335**	**4,852 / 4,850**	**57.12**	**75.49**	**69.51**	**84.58**

Table 8. Impact of Synthetic Data Augmentation: Original vs. Synthetically Expanded Data BLEU Change (5-Fold CV)

Evaluated Subset	IndicBART BLEU			mT5-Small BLEU		
	Original	Expanded	Δ	Original	Expanded	Δ
Southern Konkan (D1)	26.21	52.06	**+25.85**	48.28	**66.62**	**+18.34**
Northern Konkan (D2)	47.59	49.08	**+1.49**	62.35	**62.72**	**+0.37**
Varhadi (D4)	**72.75**	70.27	−2.48	**79.57**	78.73	−0.84
Overall Combined	**48.17**	**57.12**	**+8.95**	**63.29**	**69.51**	**+6.22**

contained garbage translations. Table 9 presents the quantitative comparison on the benchmark suite.

Key findings and insights from this evaluation include:

1. **Quality Collapse on Unverified Data**: Training IndicBART on unverified outputs causes a significant drop of 14.03 BLEU points, collapsing from 57.12 to 43.09 due to compounded error propagation from uncorrected dialect residue and wrong syntax with other errors.
2. **mT5-Small Resilience**: While google/mT5-small showed resilience due to its 250,000-token multilingual capacity, training on unverified data still suffers a 8.30 BLEU point drop ,falling from 69f.51 to 61.21 and a 4.40 chrF++ drop.
3. **Essential Role of Multi-Tier Auditing**: Removal of flawed ones and correcting the possible set of pairs is directly responsible for better performance that eliminated subsequent errors.

4.5 Model Architecture Analysis

Table 10 provides detailing on the architectural comparison between the two primary models. The 69.51 vs. 57.12 BLEU gap on the expanded multi-dialect task (+12.39 absolute) can be attributed to three key architectural differences:

1. **Vocabulary Coverage**: mT5's 250K vocabulary provides 3.9× more subword units than IndicBART's 64K, enabling finer-grained tokenization of

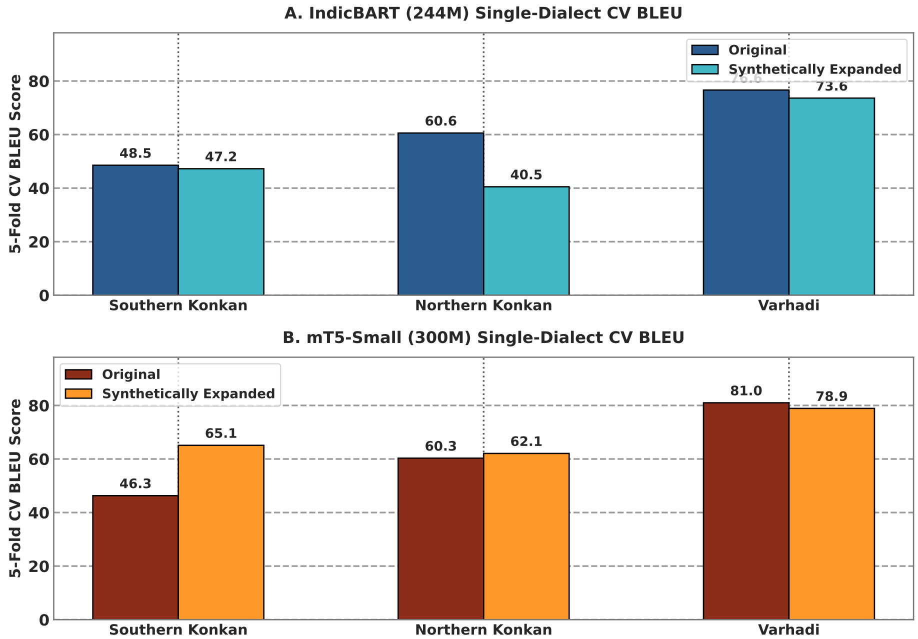

Fig. 5. Single-dialect 5-fold cross-validation performance shift comparing Original and Synthetically Expanded models for IndicBART (244M) and mT5-small (300M).

morphologically complex Marathi dialectal forms. This is critical for dialect normalization, where subtle suffix and infix changes must be captured at the subword level.

2. **Positional Encoding**: mT5 uses relative positional biases, which generalize better to variable-length sequences typical of dialectal text where sentence length may differ substantially between dialect and standard forms.
3. **Conditioning Overhead**: IndicBART requires explicit language tokens (`<2mr>`) that add conditioning complexity, whereas mT5's pure text-to-text format enables direct mapping without task-specific architectural constraints.

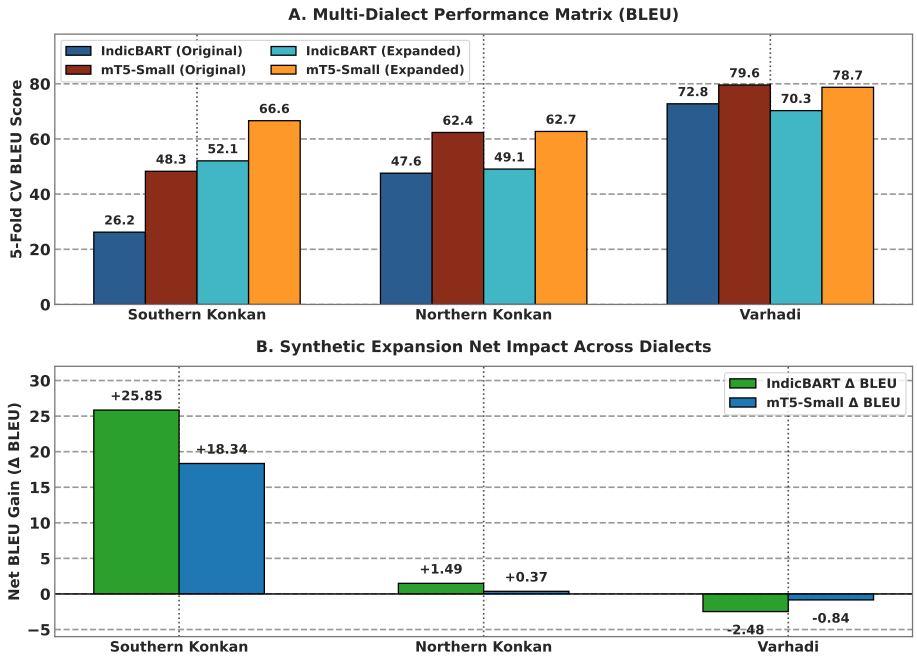

Fig. 6. Multi-dialect 5-fold cross-validation benchmark comparison: (A) Absolute BLEU scores across model architectures and corpora partitions, and (B) Net synthetic expansion impact (Δ BLEU across sub-dialects).

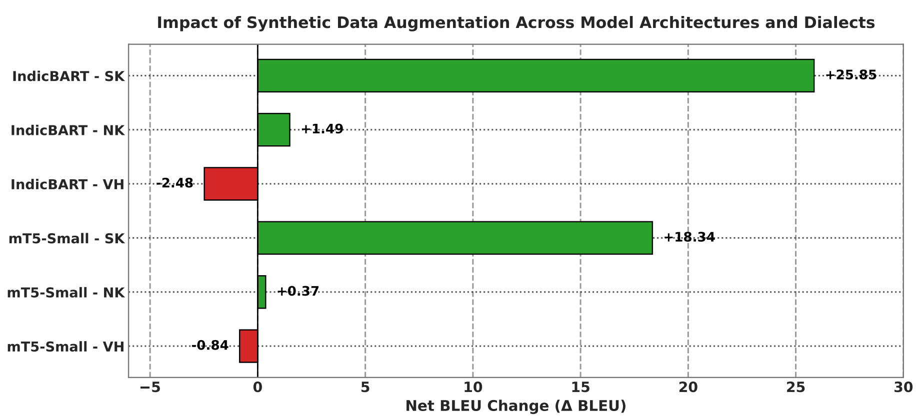

Fig. 7. Impact of LLM synthetic data expansion (Δ BLEU score comparing synthetically expanded vs. original 5-fold cross-validation benchmarks).

Table 9. Impact of Multi-Tier Verification Engine: Raw Unverified vs. Filtered Clean Synthetic Data

Model	Pipeline State	Total Pairs	BLEU	chrF++	Δ BLEU	Δ chrF++
IndicBART (244M)	Raw Unverified Data	19,914	43.09	64.54	−14.03	−10.95
	Filtered Clean Data	32,335	**57.12**	**75.49**	**+14.03**	**+10.95**
mT5-Small (300M)	Raw Unverified Data	19,914	61.21	80.18	−8.30	−4.40
	Filtered Clean Data	32,335	**69.51**	**84.58**	**+8.30**	**+4.40**

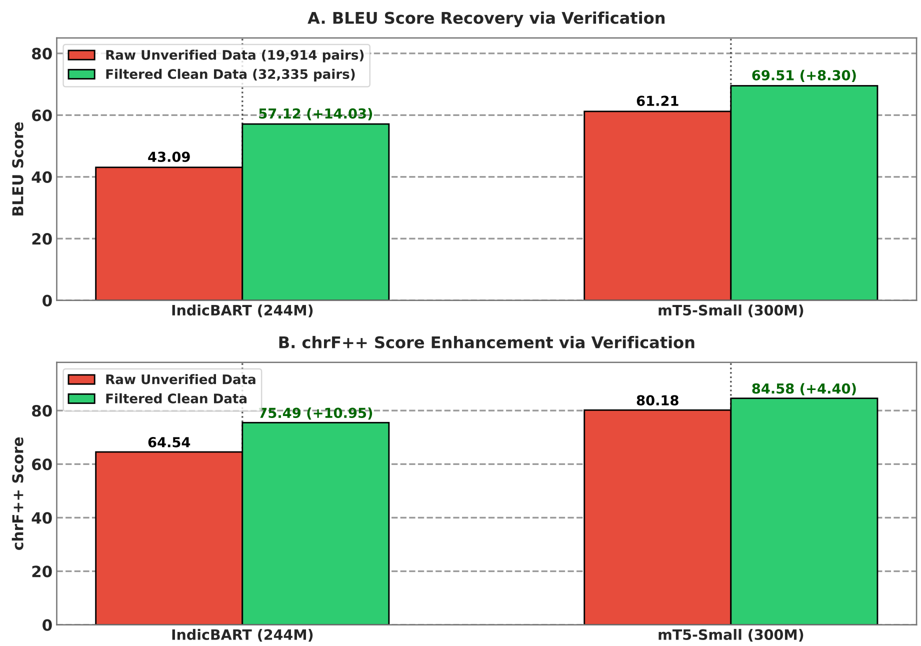

Fig. 8. Multi-tier verification engine ablation: (A) BLEU score recovery and (B) chrF++ score enhancement comparing models trained on raw unverified synthetic outputs vs. filtered clean synthetic data.

Table 10. Detailed Architecture Comparison: IndicBART (244M) vs. mT5-small (300M)

Property	IndicBART (244M)	mT5-small (300M)
Base Architecture	mBART-50 (BART Enc-Dec)	T5 Encoder-Decoder
Position Encoding	Absolute sinusoidal	Relative positional bias
Vocabulary Size	64,000 SentencePiece	250,000 SentencePiece (mC4)
Pre-training Data	11 Indic + English	101 languages (mC4)
Script Approach	Devanagari-unified	Native multi-script
Target Conditioning	Required `<2mr>` token	Raw text-to-text (no prefix)
Learning Rate	5×10^{-5} (linear warmup)	5×10^{-4} (AdaFactor)

5 Conclusion

This study presents a comprehensive, reproducible framework for multi-dialect Marathi text normalization, addressing severe data scarcity through automated synthetic data expansion and establishing rigorous neural sequence-to-sequence benchmarks.

Our LLM-driven synthetic data augmentation pipeline, powered by Gemma-4 with multi-tier verification (Devanagari script filtering and LLaMA-3.1-8B semantic auditing), successfully doubled the clean parallel corpus through synthetic expansion while maintaining strict semantic fidelity—a scalable paradigm for other low-resource dialect pairs. Fine-tuning sequence-to-sequence transformers on the expanded benchmark established state-of-the-art performance: google/mT5-small (300M) achieved 69.51 BLEU and 84.58 chrF++ on the synthetically expanded multi-dialect dataset, outperforming IndicBART (244M) by +12.39 BLEU. Individual dialect performance reached 80.99 BLEU and 91.21 chrF++ for Varhadi, demonstrating that compact pre-trained sequence-to-sequence transformers can achieve high-accuracy dialect normalization.

5.1 Limitations

This work has several limitations that should be acknowledged:

1. Synthetic data generation relies on LLM APIs (Gemma-4 via local/hosted endpoints), introducing API cost and latency considerations.
2. While we benchmarked two encoder-decoder architectures (mT5-small and IndicBART), decoder-only LLMs (e.g., LLaMA-3, Gemma-4) have not been evaluated for zero-shot or fine-tuned text normalization.
3. Evaluation is currently focused on text normalization metrics (BLEU, chrF++, WER); large-scale downstream task evaluation across diverse domain applications remains for future work.

5.2 Data and Code Availability

The complete parallel corpus of 32,335 verified dialect–Standard Marathi sentence pairs, along with all training code, evaluation scripts, and pre-trained model checkpoints, will be publicly released upon publication. The dataset will be available under an open license to support reproducible research in Indic dialect normalization. Our synthetic data generation pipeline, multi-tier verification engine, and fine-tuning configurations are documented in the accompanying code repository to enable replication and extension to additional Marathi sub-dialects and other low-resource Indic language varieties.

5.3 Future Directions

Several promising directions emerge from this work:

1. **Edge Deployment via Quantization**: Model quantization (INT8/INT4) and knowledge distillation to sub-100M parameter models for on-device deployment in rural Maharashtra.
2. **Expansion to Additional Dialects**: Extending the pipeline to Deshi/Puneri, Kolhapuri, Marathwada, and other Indic language sub-dialects.
3. **Decoder-Only LLM Benchmarking**: Evaluating instruction-tuned decoder-only models for zero-shot and few-shot dialect normalization tasks.

References

1. Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS 2017)*, pages 5998–6008, 2017.
2. Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, and et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
3. Google DeepMind Gemma Team. Gemma 4: Open language models. *arXiv preprint arXiv:2408.00118*, 2024.
4. Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. The state and fate of linguistic diversity and inclusion in the NLP world. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 6282–6293, 2020.
5. Verena Blaschke, Christoph Purschke, Hinrich Schütze, and Barbara Plank. What do dialect speakers want? a survey of attitudes towards language technology for german dialects. *arXiv preprint arXiv:2402.11968*, 2024.
6. Aditya Joshi, Raj Dabre, Diptesh Kanojia, Zhuang Li, Haolan Zhan, Gholamreza Haffari, and Doris Dippold. Natural language processing for dialects of a language: A survey. *ACM Computing Surveys*, 57(6):1–37, 2025.
7. Allison Koenecke, Andrew Nam, Emily Lake, Joe Nudell, Minnie Quartey, Zion Mengesha, Connor Toups, John R Rickford, Dan Jurafsky, and Sharad Goel. Racial disparities in automated speech recognition. *Proceedings of the National Academy of Sciences*, 117(14):7684–7689, 2020.
8. Olli Kuparinen, Aleksandra Miletić, and Yves Scherrer. Dialect-to-standard normalization: A large-scale multilingual evaluation. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 13814–13828. Association for Computational Linguistics, 2023.
9. Raj Dabre, Himani Shrotriya, Anoop Kunchukuttan, Ratish Puduppully, Mitesh M. Khapra, and Pratyush Kumar. Indicbart: A pre-trained model for indic natural language generation. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 1849–1863, 2022.
10. Nazmuddoha Ansary, Quazi Adibur Rahman Adib, Tahsin Reasat, Asif Shahriyar Sushmit, Ahmed Imtiaz Humayun, Sazia Mehnaz, Kanij Fatema, Mohammad Mamun Or Rashid, and Farig Sadeque. Unicode normalization and grapheme parsing of indic languages. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 17019–17030. ELRA Language Resource Association, 2024.

11. Niko Partanen, Mika Hämäläinen, and Khalid Alnajjar. Dialect text normalization to normative standard finnish. In *Proceedings of the 5th Workshop on Noisy User-generated Text (W-NUT @ EMNLP 2019)*, pages 141–146. Association for Computational Linguistics, 2019.
12. Saurabh Kumar, Abhayjeet Singh, Deekshitha G, Amartyaveer, Jesuraja Bandekar, Savitha Murthy, Sumit Sharma, Sandhya Badiger, Sathvik Udupa, Amala Nagireddi, Srinivasa K. M. Raghavan, Rohan Saxena, Jai Nanavati, Raoul Nanavati, Janani Sridharan, Arjun Mehta, Ashish K. S. Khuraishi, Sai Praneeth Reddy Mora, Prashanthi Venkataramakrishnan, Gauri Date, Karthika P, and Prasanta Kumar Ghosh. RESPIN-S1.0: A read speech corpus of 10000+ hours in dialects of nine indian languages. In *Advances in Neural Information Processing Systems (NeurIPS 2025) Track on Datasets and Benchmarks*, pages 1–13, 2025.
13. Yves Scherrer and Nikola Ljubešić. Normalising swiss german dialect text using character-level machine translation. In *Proceedings of the 3rd Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2016)*, pages 117–126, 2016.
14. Mathilde Lusetti, Tsetsu Ruzsics, Tanja Samardžić, and Yves Scherrer. Impact of target language models on swiss german dialect normalization. In *Proceedings of the 5th Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2018)*, pages 133–141, 2018.
15. Mihael Kresic and Noorhan Abbas. Normalizing swiss german dialects with the power of large language models. *Procedia Computer Science*, 244:287–295, 2024.
16. Mika Hämäläinen, Niko Partanen, and Khalid Alnajjar. Normalizing swedish dialects spoken in finland. In *Proceedings of the 7th Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2020)*, pages 55–62, 2020.
17. Mika Hämäläinen, Niko Partanen, and Jack Rueter. Generating dialectal finnish for dialect normalization of estonian dialects. In *Proceedings of the 9th Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2022)*, pages 71–79, 2022.
18. Kaori Abe, Kentaro Inui, and Yuji Matsumoto. Multi-dialect neural machine translation for japanese dialects. In *Proceedings of the 27th International Conference on Computational Linguistics (COLING 2018)*, pages 2105–2116, 2018.
19. Yuan Zhao and Eleanor Chodroff. Acoustic-phonetic variation in mandarin dialect corpora. *Journal of Phonetics*, 94:101172, 2022.
20. Nizar Habash, Mona Diab, and Owen Rambow. Conventional orthography for dialectal arabic. In *Proceedings of the Eighth International Conference on Language Resources and Evaluation (LREC '12)*, pages 711–718, 2012.
21. Houda Bouamor, Nizar Habash, Mohammad Salameh, et al. The madar parallel corpus of arabic dialects. In *Proceedings of the 11th Language Resources and Evaluation Conference (LREC 2018)*, pages 3375–3381, 2018.
22. Muhammad Abdul-Mageed, Chunsheng Zhang, AbdelRahim Elmadany, and Ebrahim Mohamed Nagoudi. Nadi 2023: The fourth nuanced arabic dialect identification and translation shared task. In *Proceedings of the Arabic NLP Conference (ArabicNLP @ EMNLP 2023)*, pages 512–528, 2023.
23. Fadhl Eryani, Nizar Habash, Faisal Al-Shargi, et al. A normalization corpus for five arabic dialects. In *Proceedings of the 12th Language Resources and Evaluation Conference (LREC 2020)*, pages 5750–5757, 2020.
24. Antonios Dimakis, John Pavlopoulos, and Antonios Anastasopoulos. Dialect normalization using large language models and morphological rules. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 23696–23714. Association for Computational Linguistics, 2025.

25. Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. mt5: A massively multilingual pre-trained text-to-text transformer. In *Proceedings of NAACL-HLT 2021*, pages 483–498, 2021.
26. Raviraj Joshi, Parth Goel, and Rucha Joshi. L3cube-mahanlp: Marathi natural language processing datasets, models, and library. In *Proceedings of the WILDRE-6 Workshop, LREC 2022*, pages 10–17, 2022.
27. Aman Kumar, Manish Shrivastava, and Pushpak Bhattacharyya. Indic-dialect: Benchmark dataset for indic dialect classification and translation. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 15201–15212. ELRA Language Resource Association, 2024.
28. Vineel Pratap, Ankit Saraf, and Pratyush Kumar. Wfst-guided neural text normalization for indic languages. In *Proceedings of Interspeech 2024*, pages 3120–3124, 2024.
29. Rahesha Mulla and B. Suresh Kumar. Text-independent automatic dialect recognition of marathi language using spectro-temporal characteristics of voice. *International Journal on Recent and Innovation Trends in Computing and Communication (IJRITCC)*, 10(2s):313–321, 2022.
30. Poonam S. Pawar, Jatinderkumar R. Saini, and Shraddha Vaidya. Automatic marathi dialect classification in noisy and non-noisy environments. *Procedia Computer Science*, 282:288–303, 2026.
31. Gaurav Gaikwad, Deepak Panchal, Omprasad Deshmukh, and Sonali Patil. Marathi dialect to pure marathi using rag & ai. *International Research Journal of Engineering and Technology (IRJET)*, 12(11):787–791, 2025.
32. V. A. Vyavhare, S. N. Divekar, Harshada Pawar, Karan Kale, and Subhash Jagtap. Marathi dialect translator. *International Journal of Scientific Research in Science, Engineering and Technology (IJSRSET)*, 13(3):254–261, 2026.
33. Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A. Smith, Daniel Khashabi, and Hannaneh Hajishirzi. Self-instruct: Aligning language models with self-generated instructions, 2022.
34. Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B Hashimoto. Alpaca: A strong, replicable instruction-following model. *Stanford Center for Research on Foundation Models*, 2023.
35. Subhabrata Mukherjee, Arindam Mitra, Ganesh Jawahar, Sahaj Agarwal, Hamid Palangi, and Ahmed Awadallah. Orca: Progressive learning from complex explanation traces of gpt-4, 2023.
36. Cheng-Yu Hsieh et al. Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes. *arXiv preprint arXiv:2305.02301*, 2023.
37. Branislav Pecher et al. Better as generators than classifiers: Leveraging llms and synthetic data for low-resource multilingual classification. In *Findings of the Association for Computational Linguistics: EACL 2024*, 2024.
38. Ona de Gibert et al. Scaling low-resource mt via synthetic data generation with llms. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024.
39. Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38, 2023.

40. Matt Post. A call for clarity in reporting bleu scores. In *Proceedings of WMT 2018*, pages 186–191, 2018.